

F-INVARIANT AND E-INVARIANT

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ABSTRACT. F -invariant for a pair of good elements (e.g. cluster monomials) in cluster algebras is introduced by the author in a previous work. A key feature of F -invariant is that it is a coordinate-free invariant, that is, it is mutation invariant under the initial seed mutations. E -invariant for a pair of decorated representations of quivers with potentials is introduced by Derksen, Weyman and Zelevinsky, which is also a coordinate-free invariant. The strategies used to show the mutation-invariance of F -invariant and E -invariant are totally different.

In this paper, we give a new proof of the mutation-invariance of F -invariant following the strategy used by Derksen, Weyman and Zelevinsky. As a result, we prove that F -invariant coincides with E -invariant on cluster monomials. We also give a proof of Reading's conjecture, which says that the non-compatible cluster variables in cluster algebras can be separated by the sign-coherence of g -vectors.

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1. INTRODUCTION

Cluster algebras. Cluster algebras were introduced by Fomin and Zelevinsky [FZ02] as a combinatorial approach to the upper global bases of quantum groups and to the theory of total positivity in algebraic groups. A cluster algebra of rank n (with trivial coefficients) is a $\mathbb{Z}$ -subalgebra of a rational function field generated by a special set of generators called *cluster variables*, which are grouped into overlapping subsets of fixed size n , called *clusters*. A *seed* $t = (\mathbf{x}_t, B_t)$ is a pair consisting of a cluster $\mathbf{x}_t = (x_{1;t}, \dots, x_{n;t})$ and a skew-symmetrizable integer matrix $B_t = (b_{ij}^t)_{n \times n}$, called an *exchange matrix*. One can obtain new seeds from a given one by a procedure called *mutation*. The sets of cluster variables and clusters of a cluster algebra are determined by an initial seed and the iterative mutations. A *cluster monomial* is a monomial in variables from the same cluster. The remarkable *Laurent phenomenon* [FZ02] says that the expansion of a cluster monomial u with respect to any cluster $\mathbf{x}_t$ is a Laurent polynomial.

The notions of g -vectors and F -polynomials associated to cluster monomials are introduced in [FZ07]. To each cluster monomial u and any initial seed $t = (\mathbf{x}_t, B_t)$, one has the g -vector $g_u^t \in \mathbb{Z}^n$ and F -polynomial $F_u^t(y_1, \dots, y_n) \in \mathbb{Z}[y_1, \dots, y_n]$ with respect to seed $t = (\mathbf{x}_t, B_t)$. It is known from [GHKK18] that the F -polynomial F_u^t has non-negative coefficients and constant term 1. Fomin and Zelevinsky [FZ07] proved that the Laurent expansion of u with respect to seed t can be written as:

$$(1.1) \quad u = \mathbf{x}_t^{\mathbf{g}_u^t} F_u^t(\hat{y}_{1;t}, \dots, \hat{y}_{n;t}),$$

where $\hat{y}_{k;t} = \mathbf{x}_t^{B_t \mathbf{e}_k}$ and $\mathbf{e}_k$ is the k th column of I_n . We call (1.1) the *canonical expression* of u with respect to seed t .